

Final Project Report

Smart Education Analytics and Student Performance Prediction System

Using Apache Spark, PySpark MLlib, Docker, Kubernetes, and GitHub Actions

Course	Big Data Analytics using Apache Spark
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Repository	github.com/NareshDesale/Smart_Education_Analytics
Technology Stack	Apache Spark, PySpark, MLlib, Docker, Kubernetes, GitHub Actions

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1. Introduction

Educational institutions generate large volumes of academic, attendance, online learning, examination, and placement data. Traditional processing methods become inefficient as these datasets grow in size and complexity.

This project develops a Smart Education Analytics Platform using Apache Spark to process educational datasets, generate analytical reports, perform ETL operations, and predict student placement outcomes. The solution is containerized using Docker, deployed on Kubernetes, and validated through a GitHub Actions CI/CD pipeline.

2. Problem Statement

A university requires a scalable analytics platform for student academic records, attendance data, online learning activity, examination scores, and placement records. The platform must help improve academic performance, monitor attendance, analyze subject performance, predict placement probability, and generate semester-wise reports.

3. Objectives

- Load multiple educational datasets using Apache Spark.
- Perform distributed processing using RDDs, key-value operations, caching, and persistence.
- Analyze structured data using Spark DataFrames and Spark SQL.
- Build ETL pipelines for cleaning, transforming, and storing processed data.
- Predict student placement probability using PySpark MLlib.
- Containerize the application using Docker and deploy it through Kubernetes.
- Automate build and validation through GitHub Actions.

4. System Architecture

The architecture follows a Big Data analytics pipeline: educational datasets are loaded into a SparkSession, processed through RDDs and DataFrames, queried with Spark SQL, transformed through an ETL layer, and passed to MLlib for placement prediction. The application is packaged as a Docker image, deployed through Kubernetes, and validated through CI/CD automation.

Architecture Flow: Educational datasets -> Apache Spark Session -> RDD/DataFrame processing -> Spark SQL and EDA -> ETL pipeline -> MLlib placement prediction -> Docker container -> Kubernetes deployment -> GitHub Actions CI/CD.

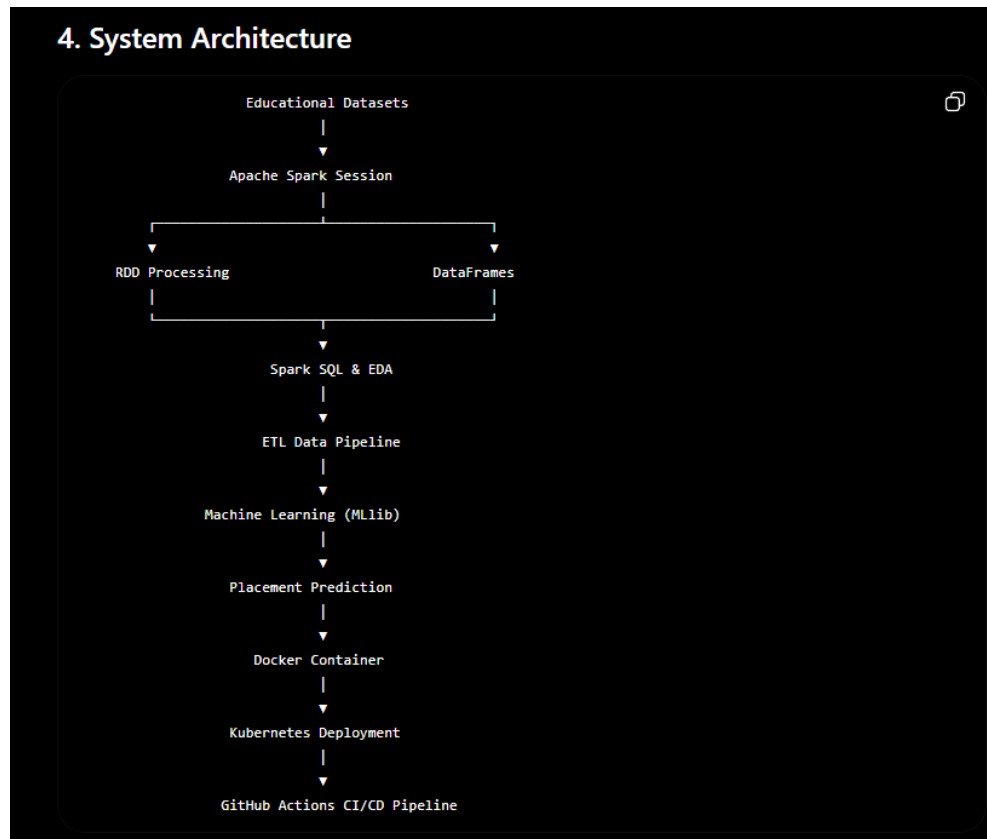


Figure 1. System architecture showing educational datasets, Apache Spark processing, Spark SQL and EDA, ETL pipeline, MLlib placement prediction, Docker, Kubernetes, and CI/CD.

5. Technology Stack

Category	Technology
Programming Language	Python
Big Data Framework	Apache Spark
Spark API	PySpark
SQL Engine	Spark SQL
Machine Learning	PySpark MLlib
Containerization	Docker
Orchestration	Kubernetes
Version Control	Git and GitHub
CI/CD	GitHub Actions

Category	Technology
IDE	Visual Studio Code

6. Datasets Used

Dataset	Purpose
xAPI Educational Dataset	Attendance analysis, student engagement, and learning activity analysis
Student Mathematics Dataset	Academic performance and mathematics score analysis
Student Portuguese Dataset	Subject-wise performance analysis
Campus Placement Dataset	Placement prediction and placement trend analysis

7. Project Folder Structure

The repository is organized into dedicated folders for source code, data, notebooks, documentation, Kubernetes manifests, screenshots, and CI/CD workflow configuration.

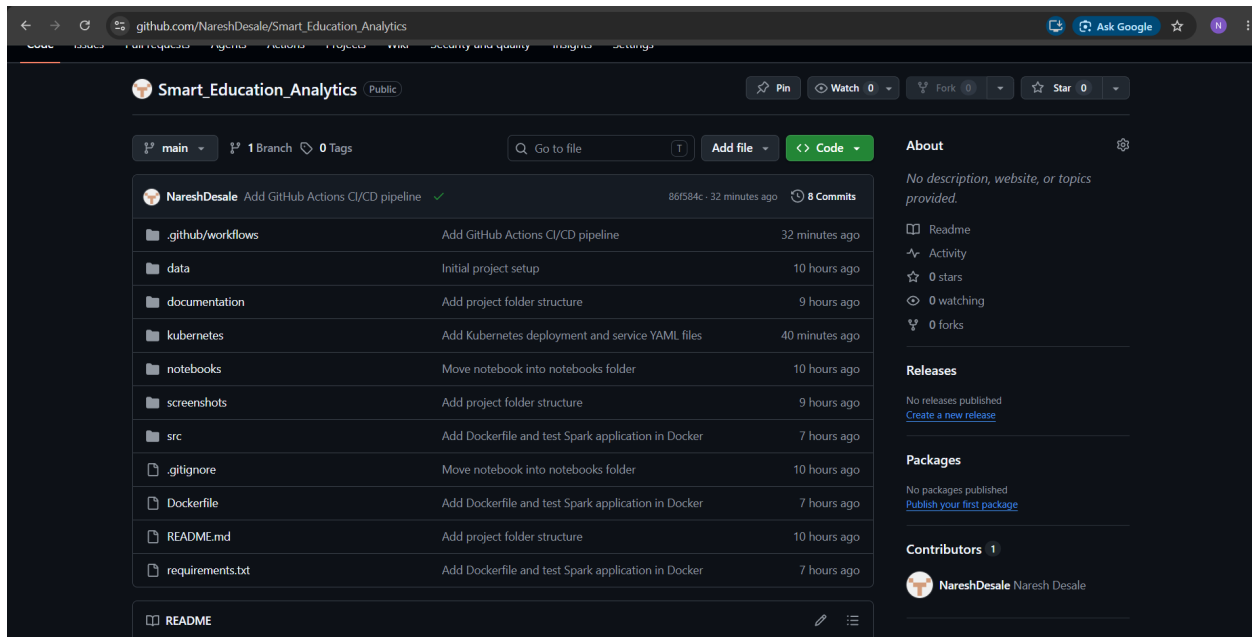


Figure 2. GitHub repository structure showing data, notebooks, source code, Dockerfile, Kubernetes manifests, and GitHub Actions workflow.

8. Implementation

8.1 Spark Initialization

A SparkSession is created, educational datasets are loaded, and schemas are validated before analytical processing.

8.2 RDD and Key-Value Processing

RDD operations such as map, filter, collect, count, and take support distributed processing of educational records. Key-value transformations including reduceByKey, groupByKey, sortByKey, caching, and persistence enable efficient aggregation.

8.3 DataFrame and Spark SQL Analytics

DataFrame filtering, joins, grouping, and aggregation are used for structured analysis. Spark SQL queries support attendance pattern analysis, subject-wise performance, top-performing students, placement trends, and semester-wise reports.

8.4 ETL Pipeline

- 1 Extract educational datasets.
- 2 Transform records by removing duplicates, handling missing values, renaming columns, and cleaning data.
- 3 Load processed data as Parquet files for efficient downstream analytics.

8.5 Machine Learning

A Logistic Regression model is used to predict student placement probability. The model is evaluated using accuracy, precision, recall, and F1 score.

9. Docker Containerization

The Spark application is containerized through Docker. The Dockerfile installs Python, Java, Spark dependencies, project requirements, and application code. The built image can execute the PySpark application consistently across environments.

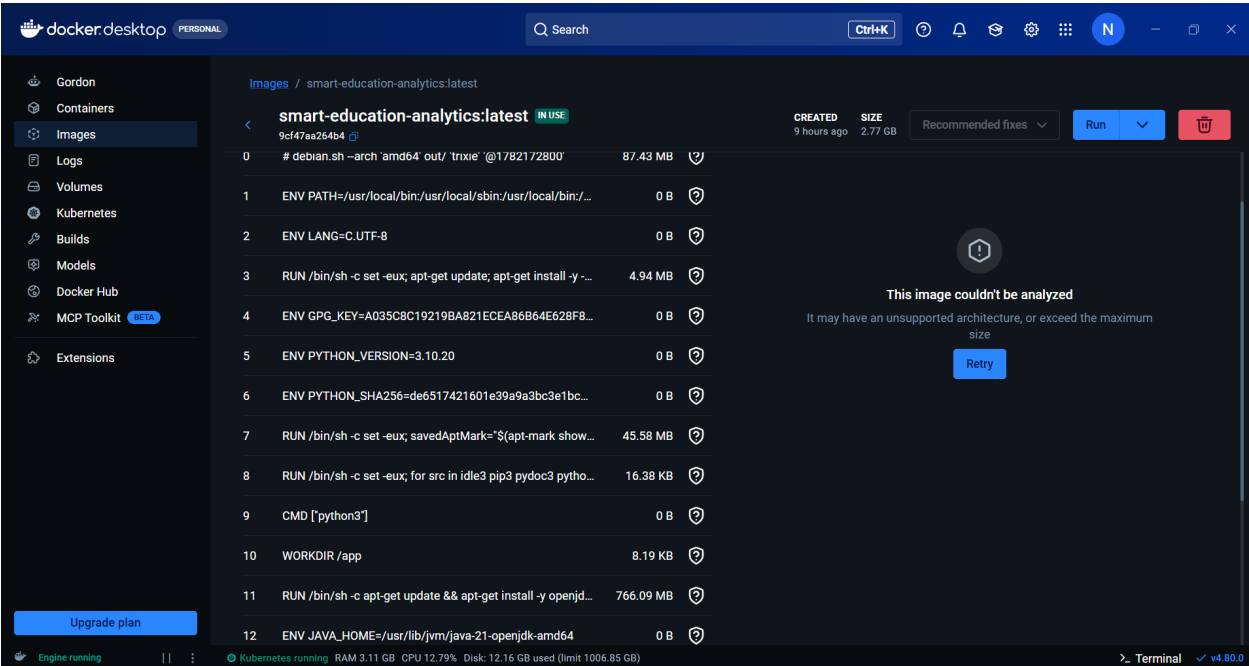


Figure 3. Docker Desktop view for the smart-education-analytics:latest image, showing image size and active usage.

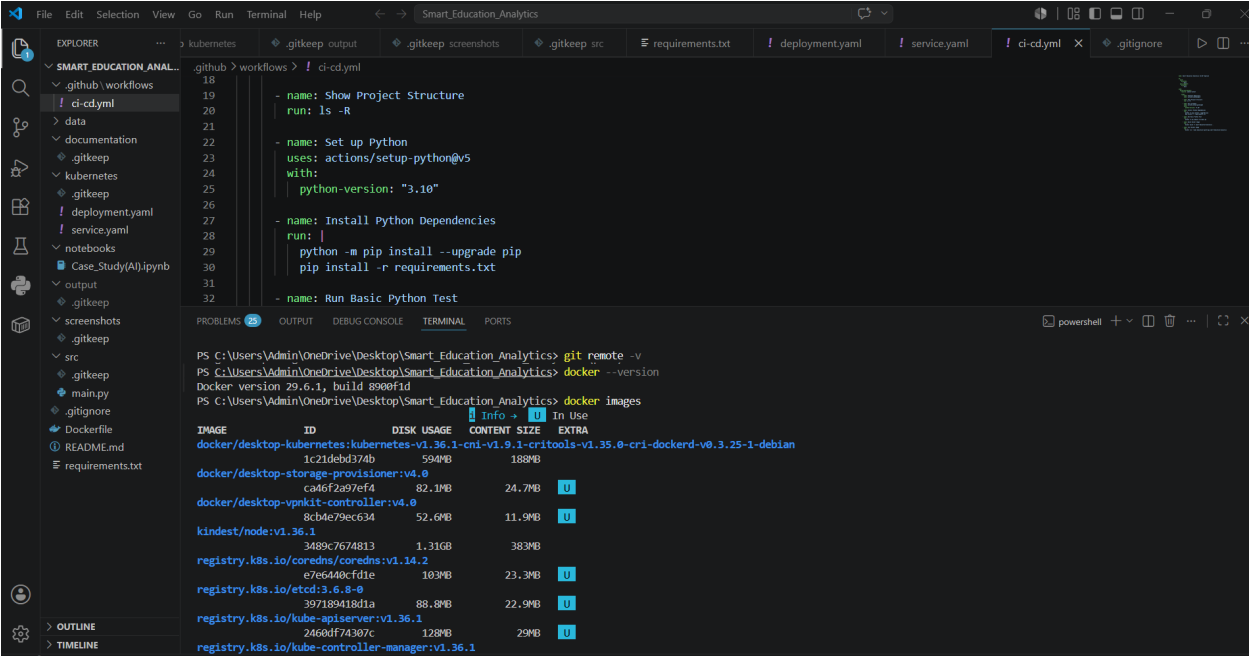


Figure 4. Docker CLI verification showing Docker version and available Kubernetes-related images.

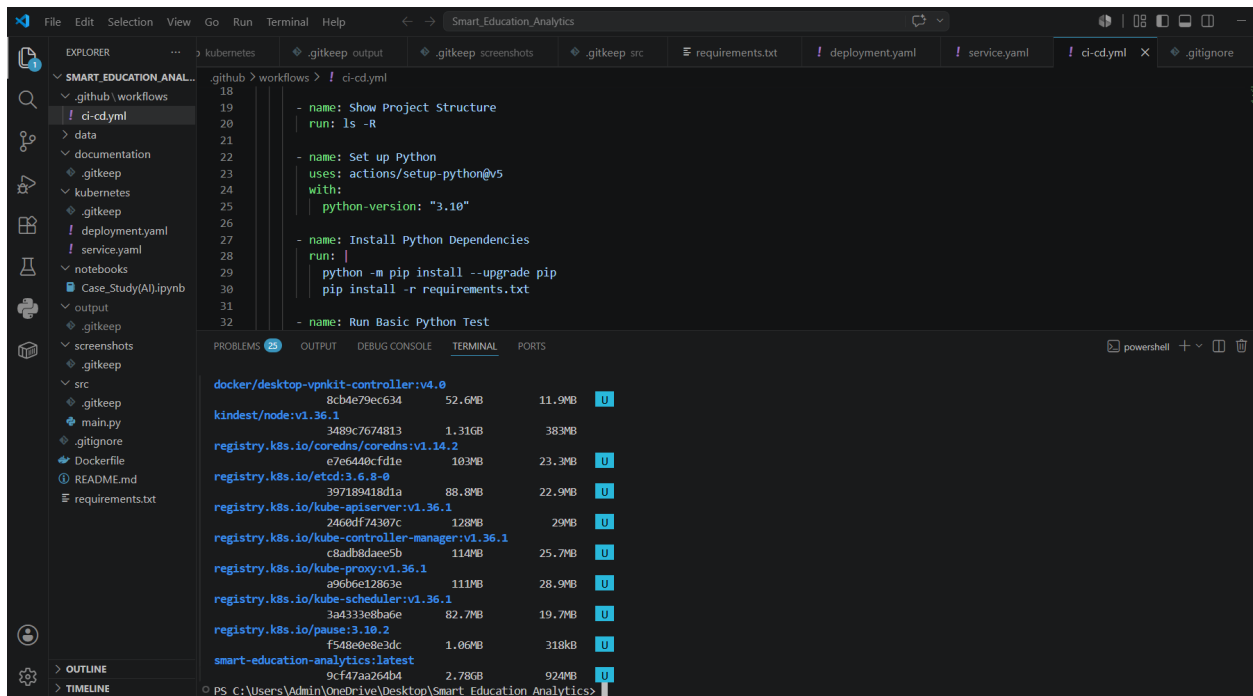


Figure 5. Docker image list showing smart-education-analytics:latest successfully built.

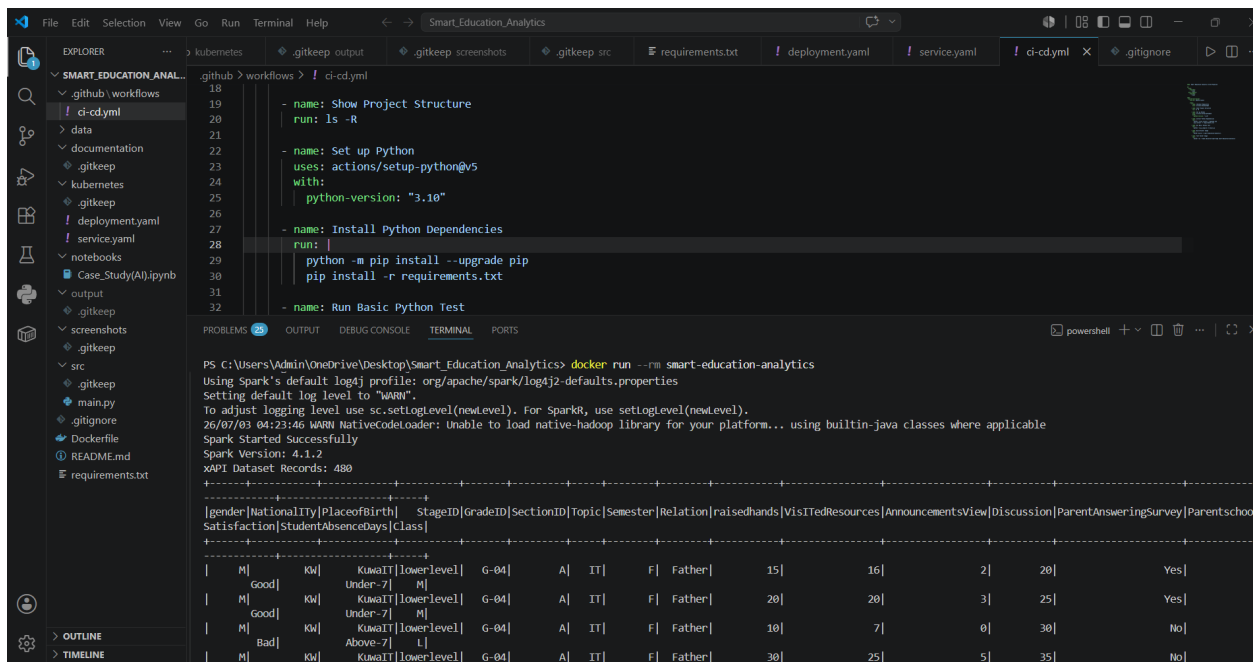


Figure 6. Docker run output showing Spark startup, Spark version 4.1.2, and xAPI dataset record count of 480.


```

github > workflows > ! ci-cd.yml
18
19 - name: Show Project Structure
20   run: ls -R
21
22 - name: Set up Python
23   uses: actions/setup-python@v5
24   with:
25     python-version: "3.10"
26
27 - name: Install Python Dependencies
28   run: |
29     python -m pip install --upgrade pip
30     pip install -r requirements.txt
31
32 - name: Run Basic Python Test

```

```

PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> docker run --rm smart-education-analytics
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|gender|Nationality|PlaceofBirth| StageID|GradeID|SectionID|Topic|Semester|Relation|raisedhands|visitedResources|AnnouncementsView|Discussion|ParentAnsweringSurvey|Parentscho
Satisfaction|StudentAbsenceDays|Class|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| M| Good| Kw| KuwaIT|Lowerlevel| G-04| A| IT| F| Father| 15| 16| 2| 20| Yes|
| M| Good| Kw| KuwaIT|Lowerlevel| G-04| A| IT| F| Father| 20| 20| 3| 25| Yes|
| M| Good| Kw| KuwaIT|Lowerlevel| G-04| A| IT| F| Father| 10| 7| 0| 30| No|
| M| Bad| Kw| KuwaIT|Lowerlevel| G-04| A| IT| F| Father| 30| 25| 5| 35| No|
| M| Bad| Kw| KuwaIT|Lowerlevel| G-04| A| IT| F| Father| 40| 50| 12| 50| No|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics>

```

Figure 7. Docker run output displaying the top five records from the educational dataset.

10. Kubernetes Deployment

Kubernetes deployment and service manifests are created to run the containerized Spark application. Deployment verification confirms the Docker Desktop node is ready, the application deployment is available, the pod is running, and the NodePort service exposes the application.

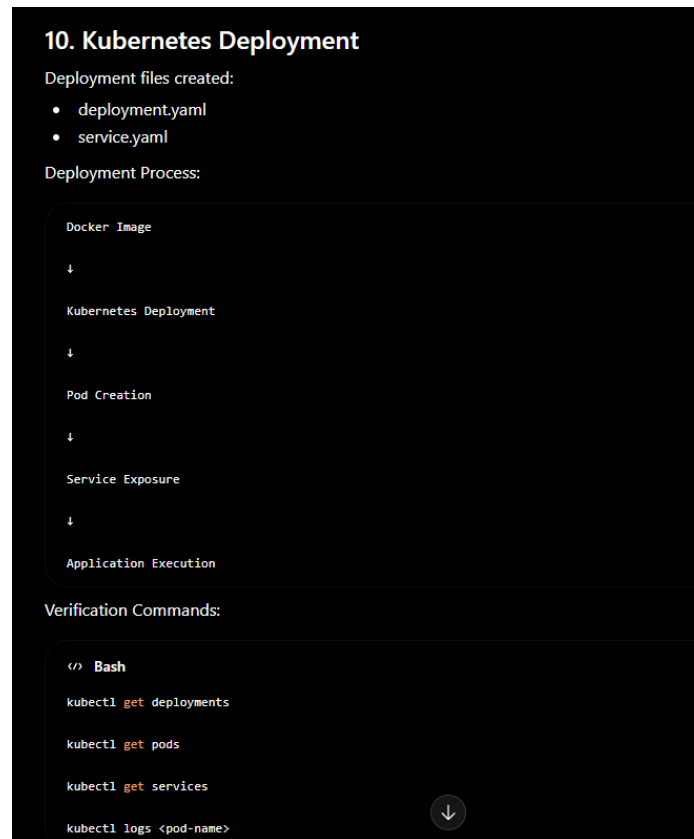


Figure 8. Kubernetes deployment architecture showing Docker image creation, deployment, pod creation, service exposure, and application execution.

```

PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> docker run --rm smart-education-analytics
Bad| Above-7| L|
M| Kw| KuwaIT|lowerlevel| G-04| A| IT| F| Father| 40| 50| 12| 50| No|
Bad| Above-7| M|

only showing top 5 rows
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
docker-desktop      Ready    control-plane   65m   v1.36.1
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get deployments
NAME                READY    UP-TO-DATE    AVAILABLE    AGE
smart-education-deployment  1/1      1              1            56m
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get pods
NAME                READY    STATUS    RESTARTS    AGE
smart-education-deployment-5694984f99-z2kz2  1/1      Running    14 (2m18s ago)  57m
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1     <none>          443/TCP    65m
smart-education-service  NodePort    10.101.2.57   <none>          80:30329/TCP  57m
  
```

Figure 9. Kubernetes verification showing ready node, available deployment, running pod, and NodePort service.

11. DevOps CI/CD Pipeline

The GitHub Actions workflow automates project validation by checking out the repository, displaying project structure, setting up Python, installing dependencies, running a basic Python test, building the Docker image, and testing the Docker image.

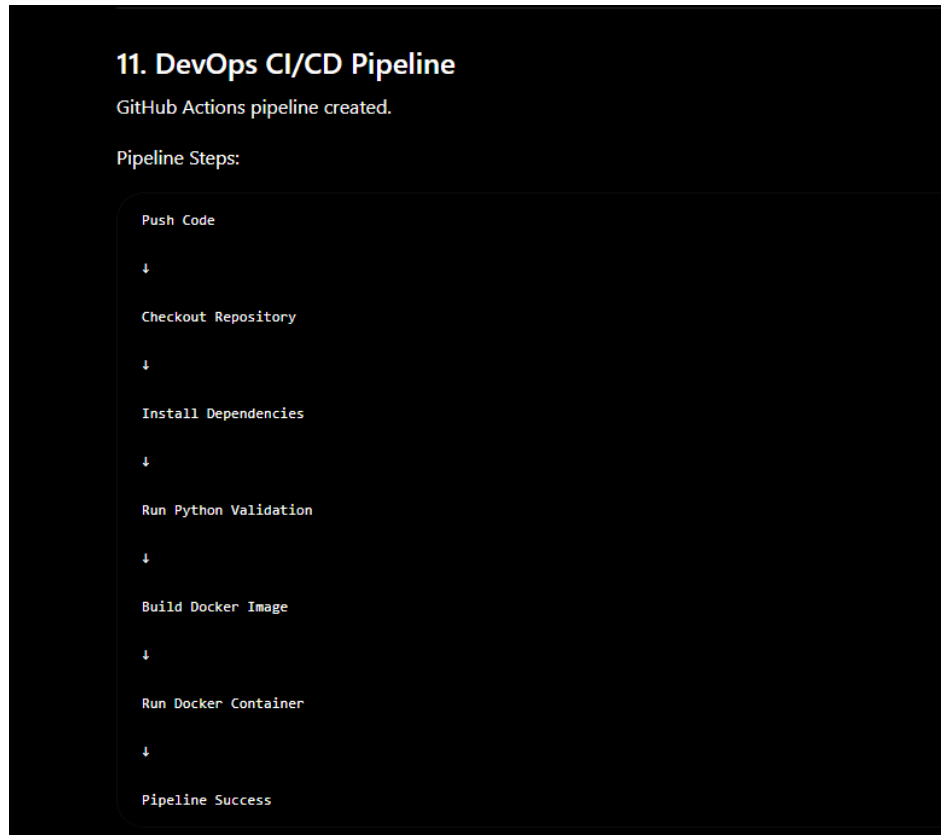


Figure 10. DevOps CI/CD pipeline architecture showing push, checkout, dependency installation, Python validation, Docker build, container run, and pipeline success.

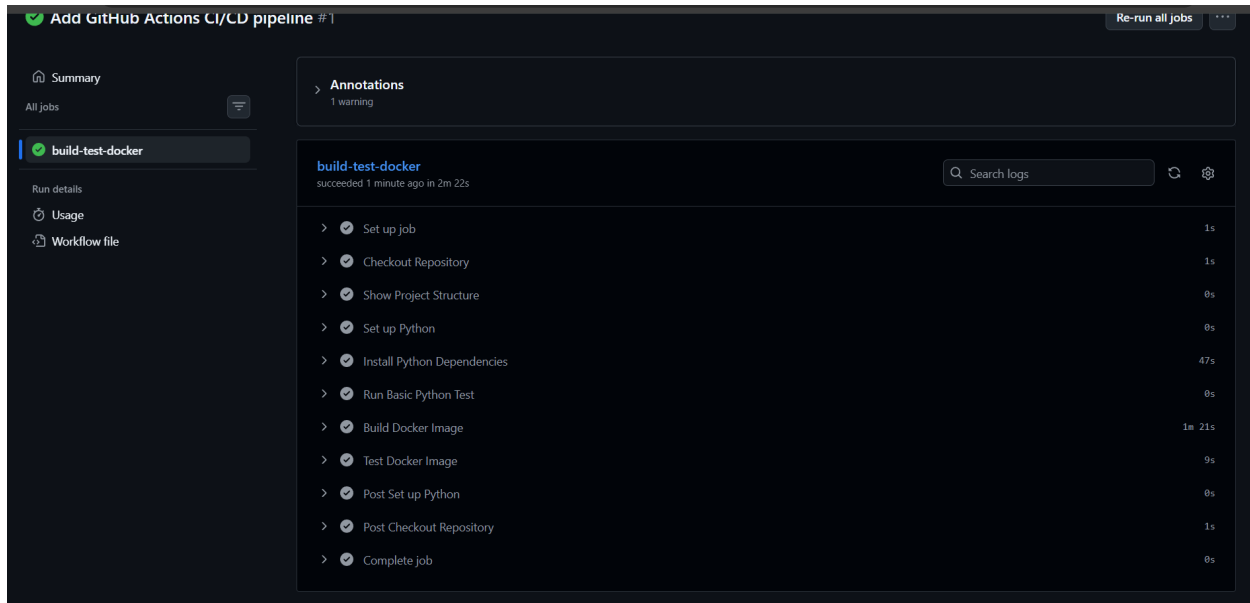


Figure 11. GitHub Actions pipeline run showing the build-test-docker job completed successfully in 2 minutes and 22 seconds.

12. Results and Output Evidence

The project successfully processes educational datasets using Apache Spark, performs RDD and DataFrame operations, generates attendance and academic analytics, executes the ETL pipeline, predicts student placement, and validates the application through Docker, Kubernetes, and GitHub Actions.

Validation Area	Observed Result
Spark execution	Spark started successfully; Spark version 4.1.2 displayed during container execution.
Dataset loading	xAPI dataset loaded with 480 records and sample rows displayed from the container.
Docker image	smart-education-analytics:latest image built successfully and visible in Docker CLI/Desktop.
Kubernetes	Deployment ready 1/1, pod running 1/1, and NodePort service exposed on port 30329.
CI/CD	GitHub Actions build-test-docker workflow succeeded, including dependency installation and Docker validation.

Result Summary: The outputs confirm that the analytics code, container image, Kubernetes deployment, and CI/CD pipeline are functioning as an integrated Big Data project workflow.

13. Conclusion

This project demonstrates a complete educational analytics solution using Apache Spark. It integrates distributed data processing, Spark SQL, ETL pipelines, machine learning, Docker containerization, Kubernetes orchestration, and DevOps automation.

The platform enables educational institutions to analyze academic performance efficiently and predict student outcomes using scalable Big Data technologies.

14. Future Scope

- Real-time analytics using Spark Streaming.
- Apache Kafka integration for live data ingestion.
- Cloud deployment on AWS, Azure, or Google Cloud Platform.
- Interactive dashboard using Streamlit.
- Advanced deep learning models for prediction.
- Student recommendation system and AI-powered academic advisor.

15. GitHub Repository

Repository Name: Smart_Education_Analytics

Repository Link: https://github.com/NareshDesale/Smart_Education_Analytics